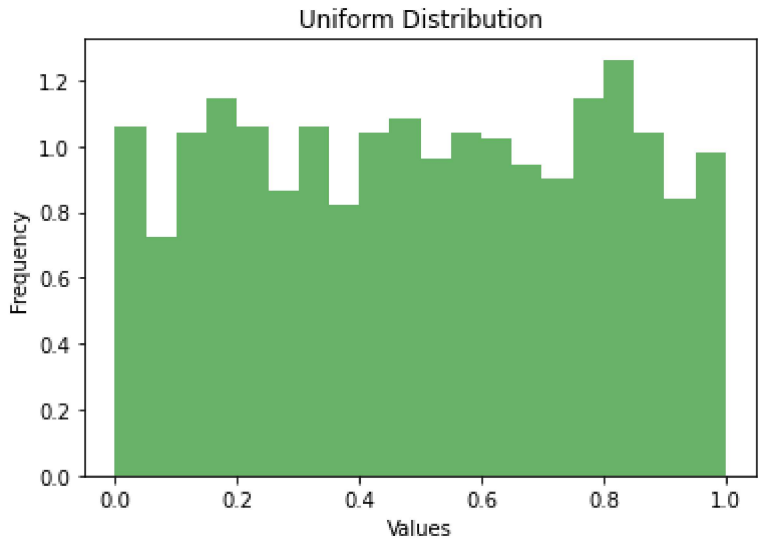
Uniform Distribution

In [7]:

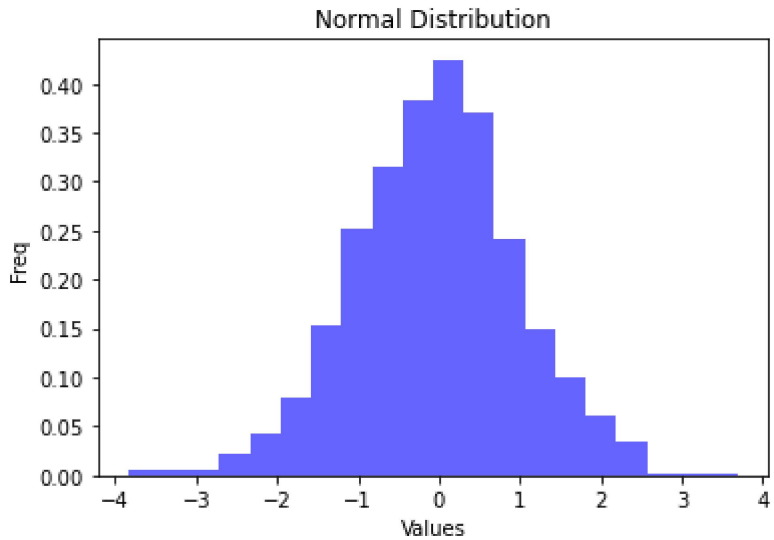
```
import matplotlib.pyplot as plt
import numpy as np

data=np.random.uniform(0,1,1000)
plt.hist(data, bins=20, density=True, alpha=0.6, color='g')
plt.title("Uniform Distribution")
plt.xlabel("Values")
plt.ylabel("Frequency")
plt.show()
```



Normal Distribution

In [6]:

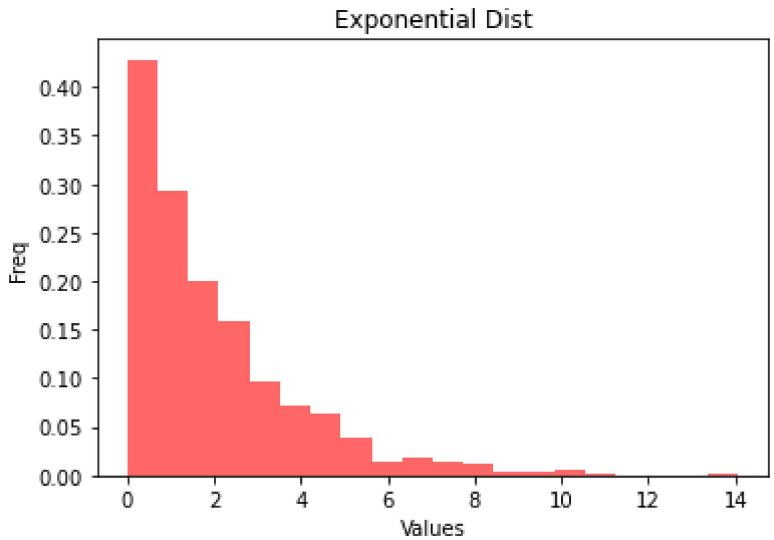
```
data = np.random.normal(0,1,1000)
plt.hist(data, bins = 20, density = True, alpha = 0.6, color = 'b')
plt.title("Normal Distribution")
plt.xlabel("Values")
plt.ylabel("Freq")
plt.show()
```



Exponential Distribution

In [11]:

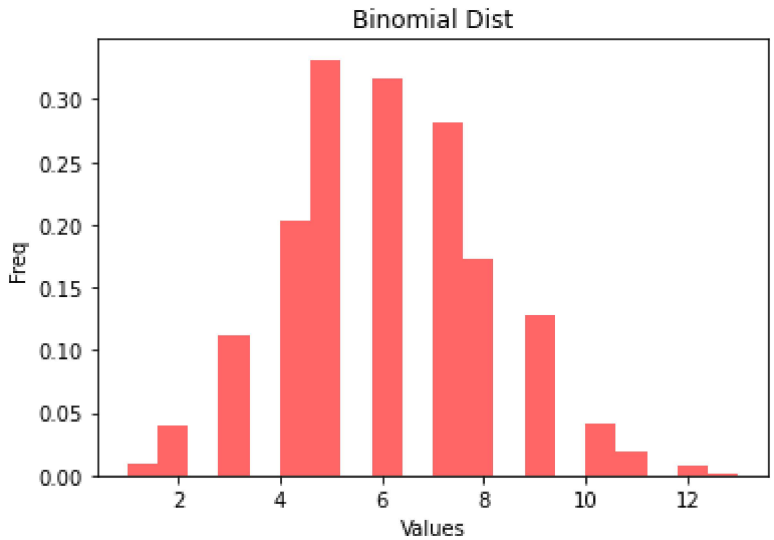
```
data = np.random.exponential(1/0.5, 1000)
plt.hist(data, density = True, bins=20, alpha=0.6, color = 'r')
plt.title("Exponential Dist")
plt.xlabel('Values')
plt.ylabel('Freq')
plt.show()
```



Binomial Distribution

In [14]:

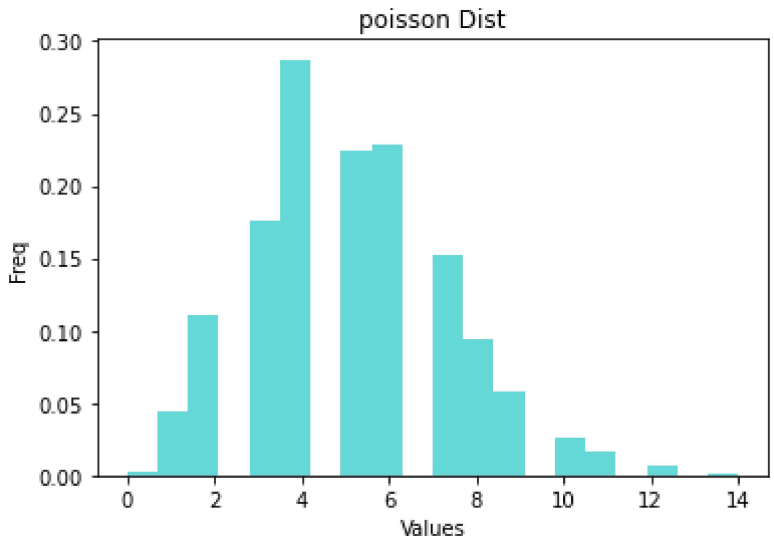
```
data = np.random.binomial(20, 0.3, 1000)
plt.hist(data, density = True, bins=20, alpha=0.6, color = 'r')
plt.title("Binomial Dist")
plt.xlabel('Values')
plt.ylabel('Freq')
plt.show()
```



Poisson Distribution

In [17]:

```
data = np.random.poisson(5, 1000)
plt.hist(data, density = True, bins=20, alpha=0.6, color = 'c')
plt.title("poisson Dist")
plt.xlabel('Values')
plt.ylabel('Freq')
plt.show()
```



In [ ]:

```

```